

$$\frac{dz}{dt} = \frac{dx}{dn} \times \frac{dn}{dt} + \frac{dx}{dy} \times \frac{dy}{dt}$$

$$\frac{dz}{dt} = (ny+1)e^{ny} \times \left[\frac{2}{t} \right] + n^2 e^{ny} \left[-\frac{1}{t^2} \right]$$

Ex₁ From Book

⇒ Use the chain Rule to find the derivative of

$w = ny$
with respect to t along the path
 $n = \cos t$, $y = \sin t$, what is the derivative's value at $t = \pi/2$?

sol

we apply the chain Rule to find $\frac{dw}{dt}$ as follows:-

$$\frac{dw}{dt} = \frac{dw}{dn} \frac{dn}{dt} + \frac{dw}{dy} \frac{dy}{dt}$$

Now Replace the values:-

$$\frac{dw}{dt} = \frac{d(ny)}{dn} \cdot \frac{d(\cos t)}{dt} + \frac{d(ny)}{dy} \frac{d(\sin t)}{dt}$$

$$= (y)(-\sin t) + n(\cos t)$$

$$= (\sin t)(-\sin t) + (\cos t)(\cos t)$$

$$= -\sin^2 t + \cos^2 t$$

$$= \cos 2t \quad \text{Now}$$

at the given value of t .

$$\left(\frac{dw}{dt}\right)_{t=\frac{\pi}{2}} = \cos\left(\frac{2 \times \pi}{2}\right) = \cos \pi = -1$$

Ans

⇒ Functions of three variables:-

If $w = (x, y, z)$ is differentiable and x, y , and z are differentiable functions of t . then w is a differentiable function of t and.

$$\frac{dw}{dt} = \frac{dw}{dx} \frac{dx}{dt} + \frac{dw}{dy} \frac{dy}{dt} + \frac{dw}{dz} \frac{dz}{dt}$$

→ ex: 21

find $\frac{dw}{dt}$ if

$$w = xy + z, \quad x = \cos t, \quad y = \sin t, \quad z = t$$

$$\left[\frac{dw}{dt} = \frac{dw}{dx} \frac{dx}{dt} + \frac{dw}{dy} \frac{dy}{dt} + \frac{dw}{dz} \frac{dz}{dt} \right]$$

$$= (y) (-\sin t) + (x) (\cos t) + (1)(1)$$

$$= (\sin t) (-\sin t) + (\cos t) (\cos t) + 1$$

$$= -\sin^2 t + \cos^2 t + 1 =$$

$$\rightarrow \text{trigonometric identity: } \cos^2 t - \sin^2 t = \cos 2t$$

$$1 + \cos 2t$$

$$\left(\frac{dw}{dt}\right)_{t=0} = 1 + \cos(0) = 2$$

Ans